Q2. **a)** Find I_1, I_2, I_3, V_1, V_2 & V_0 in the following circuit. [5]

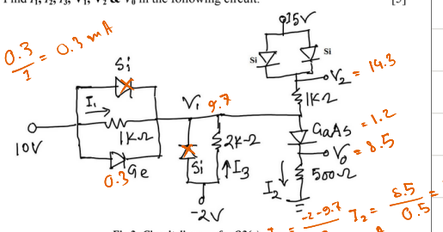


Fig 2: Circuit diagram for Q2(a)

$I_1 = \frac{0.3}{1} = 0.3$ mA

$V_2 = 19.3$

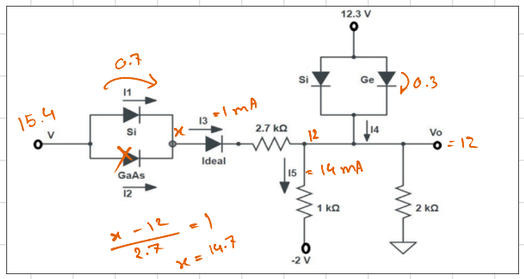
$V_1 = 9.7$

$V_0 = 8.5$

$W = \frac{2\pi}{T}$

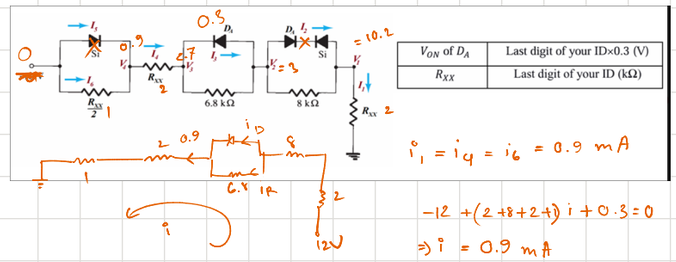
$I_3 = \frac{-2 - 9.7}{2} = -5.85$ mA

$I_2 = \frac{8.5}{0.5} = 17$ mA



$\frac{x - 12}{2.7} = 1$

$x = 14.7$



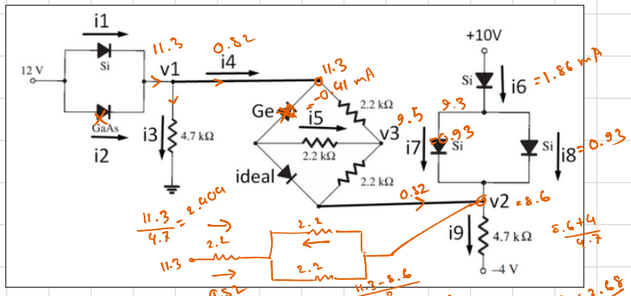
V_{ON} of D_A	Last digit of your ID×0.3 (V)
R_{XX}	Last digit of your ID (kΩ)

$i_1 = i_4 = i_6 = 0.9$ mA

$-12 + (2+8+2+1)\,i + 0.3 = 0$

$\Rightarrow i = 0.9$ mA

$i_R = \frac{0.3}{6.8} = 0.044$ mA $\quad i_D = 0.856$ mA



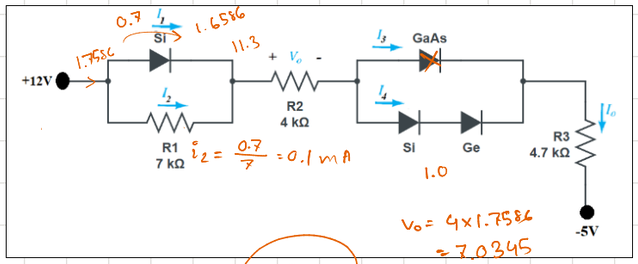
$\frac{11.3}{4.7} = 2.404$

$\frac{11.3 - 8.6}{3.3} = 0.82$

$i_6 + 0.82 = 2.68$

$i_6 = 1.86$

$\frac{8.6 + 4}{4.7}$

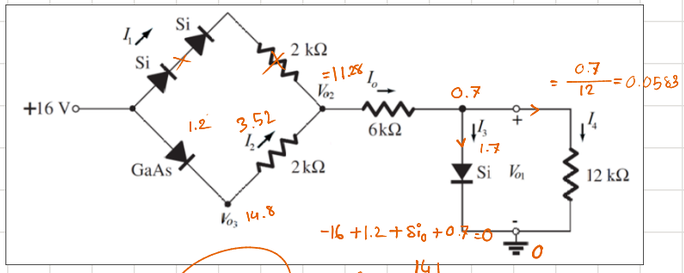


$i_2 = \frac{0.7}{7} = 0.1$ mA

$V_o = 4 \times 1.7586 = 7.0345$

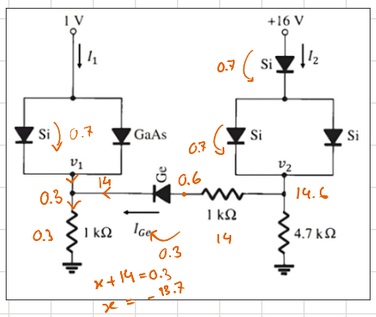
$-12 + 0.7 + (4 + 4.7)\,i + 1 - 5 = 0$

$\Rightarrow i = \frac{10.3}{8.7} = 1.7586$ mA



$= \frac{0.7}{12} = 0.0583$

$i_o \Rightarrow i_o = \frac{14.1}{8} = 1.76$ mA



$x + 14 = 0.3$

$x = -13.7$